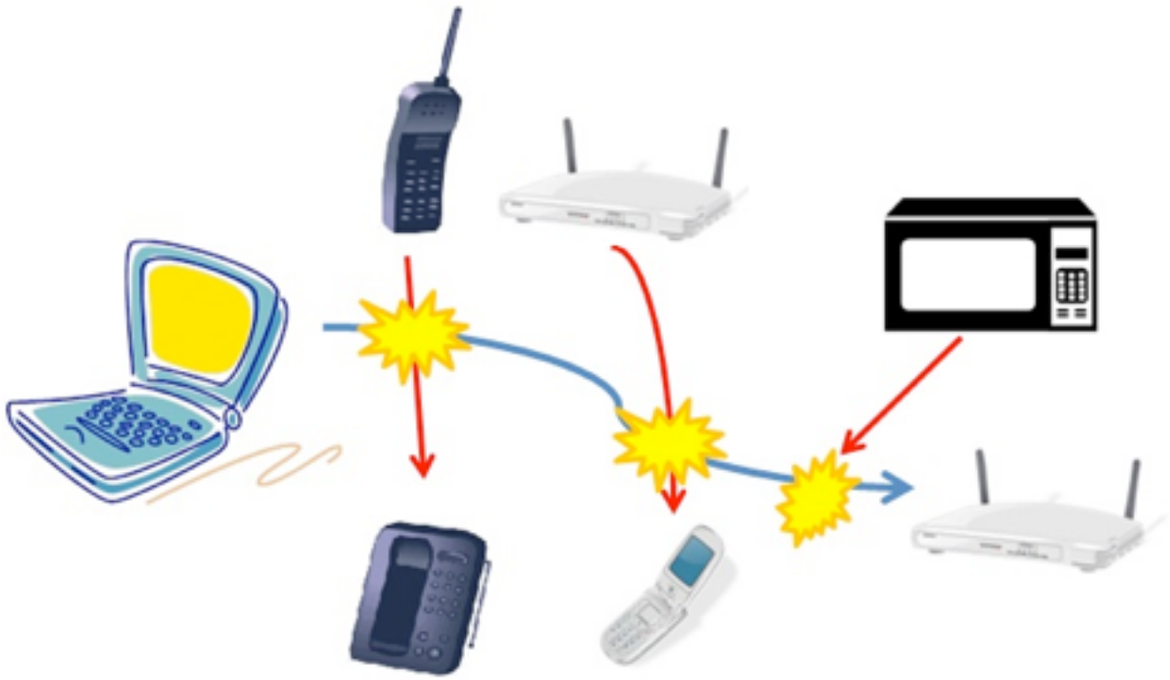
„Minstrel-Blues"
Practical Joint Rate und Power Control in Linux mac80211

Thomas Hühn

TU-Berlin

Problem



Wireless network capacity is bounded by interference

Can we increase the efficiency of WiFi spectrum usage ?

How to manage interference to increase spatial reuse ?

Can we fulfill regulatory TPC requirements (802.11h) ?

2

WiFi Resource Allocation



IEEE 802.11

Link scheduling
Rate control

Carrier sense control
Power control (TPC)

@MAC layer

cross-layer ?

@physical layer

Status quo:

todays APs use a static (max) power level to all STAs

Working parts of Minstrel-Blues



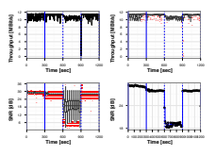
- Linux mac80211 extension to enable power level annotations in sample packets and rate-power table



- Design and implementation of decentralized joint power & rate control per packet: “Minstrel-Blues”
  - working with Minstrel & ath5k & ath9k

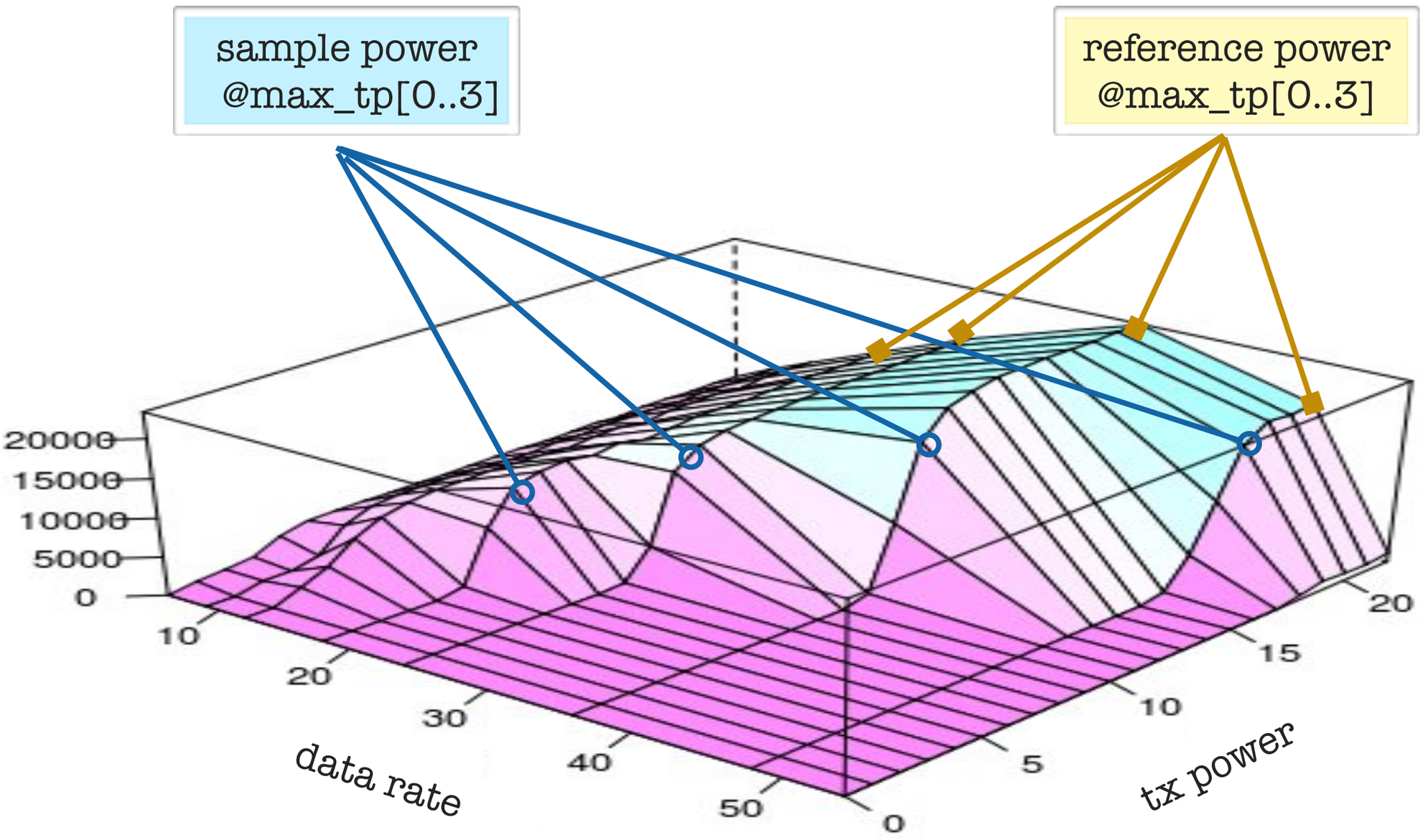
- global Ack Power control
- new mac80211 driver flags (TPC per packet, per mrr, ACK)

- Weight your interference preference with a utility function

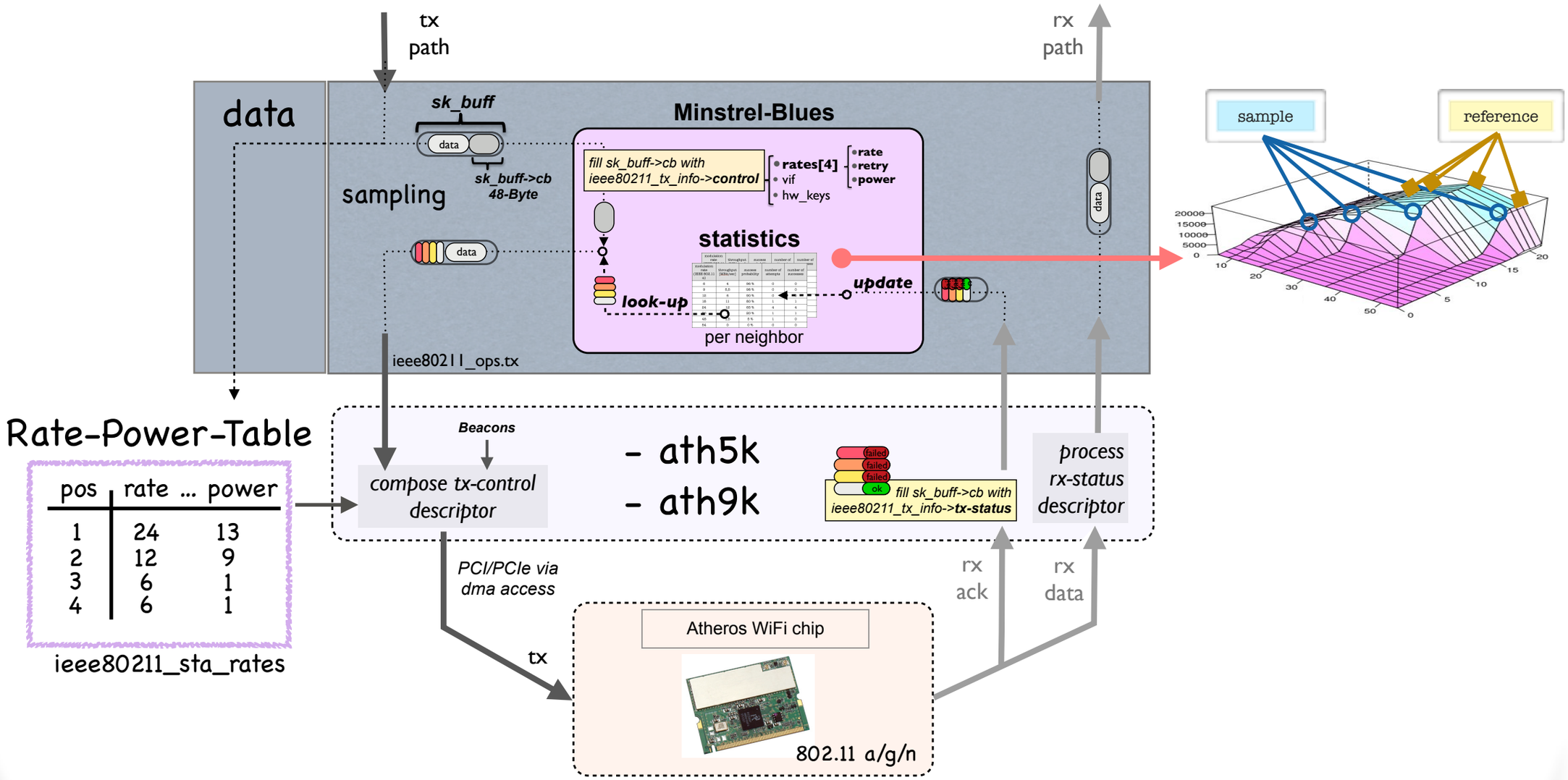


- Validation and Performance Analysis

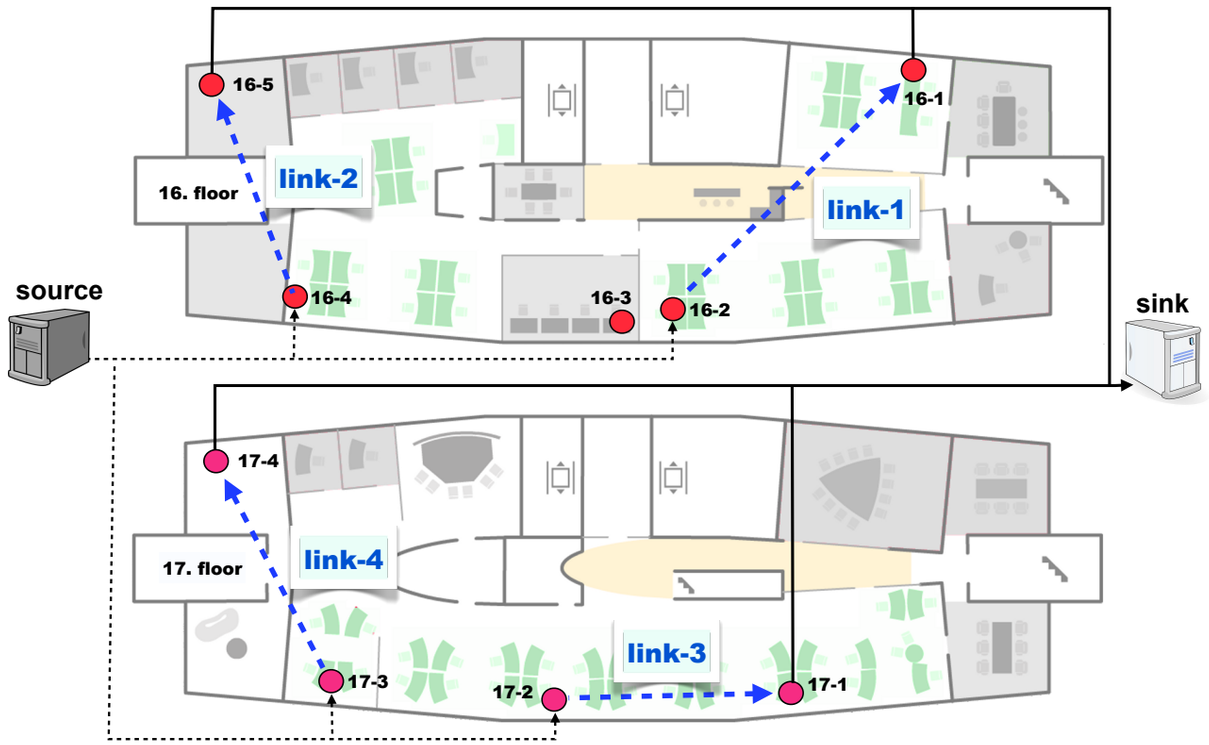
Throughput as Function (rate,power)

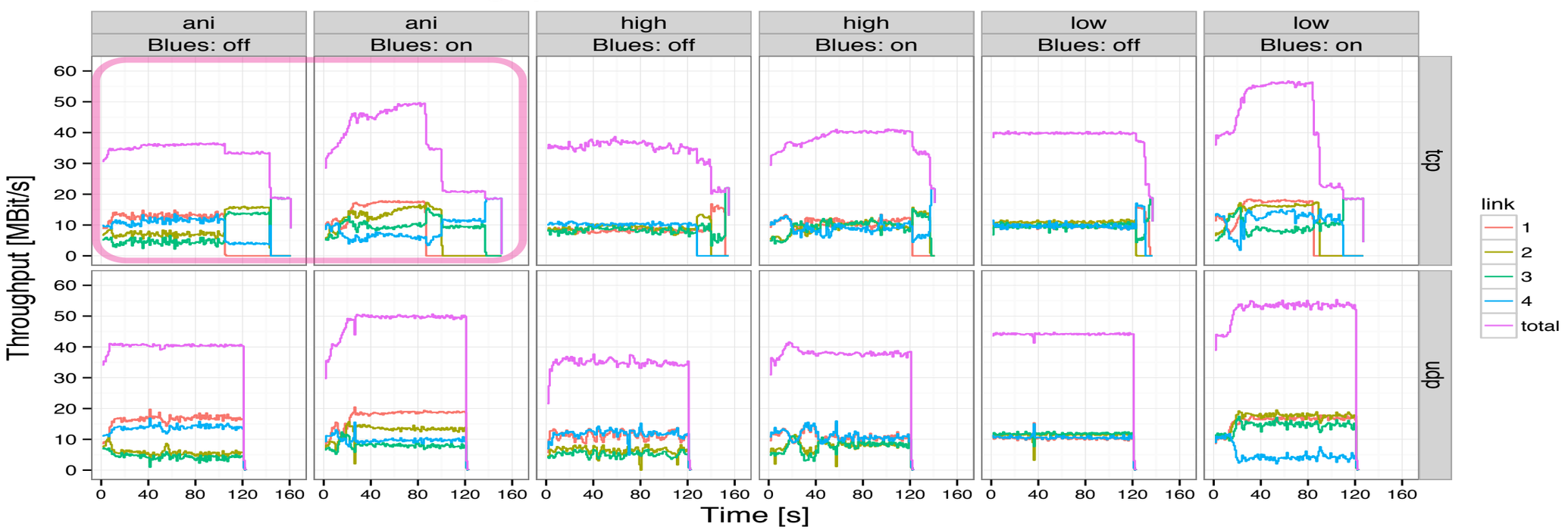


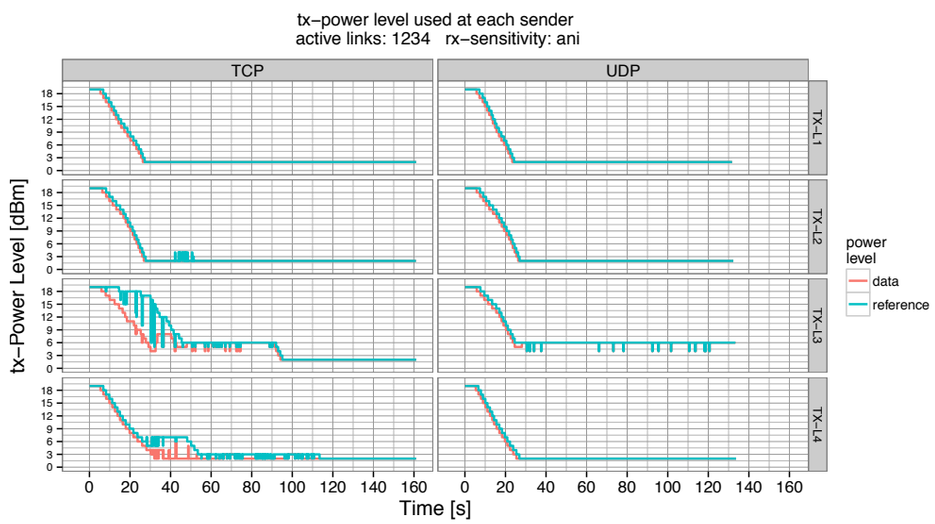
new rate-power-table

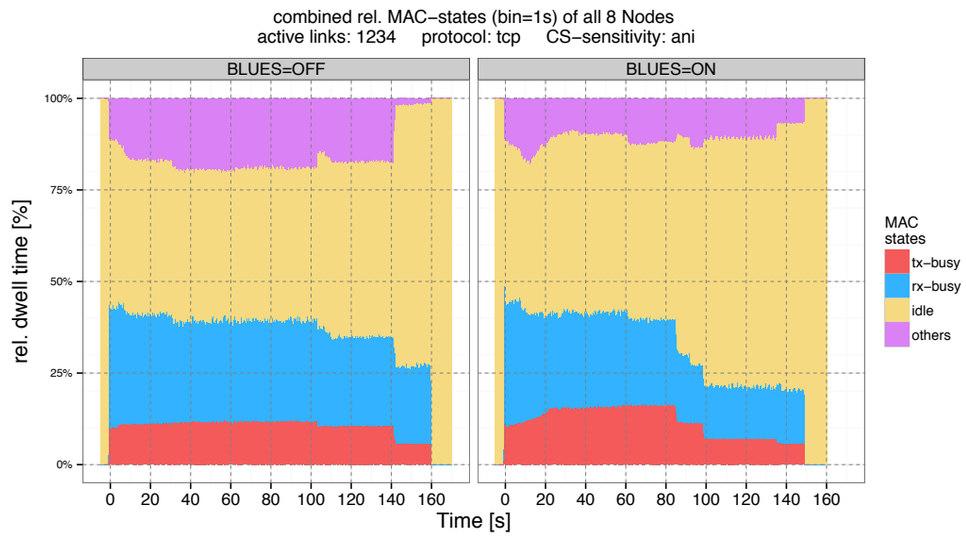


Performance with 4 active indoor links




Throughput with active links: 1, 2, 3, 4
ani
Blues: off
ani
Blues: on
high
Blues: off
high
Blues: on
low
Blues: off
low
Blues: on
tcp
udp
Throughput [MBit/s]
0
10
20
30
40
50
60
0
40
80
120
160
Time [s]
link
1
2
3
4
total

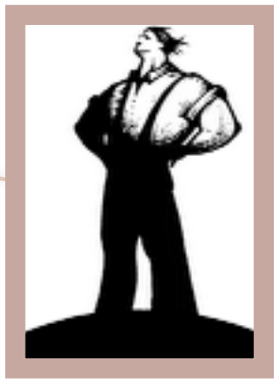

tx−power level used at each sender
active links: 1234 rx−sensitivity: ani
TCP
UDP
TX−L1
TX−L2
TX−L3
TX−L4
tx−Power Level [dBm]
0
3
6
9
12
15
18
0
20
40
60
80
100
120
140
160
Time [s]
power level
data
reference


combined rel. MAC−states (bin=1s) of all 8 Nodes
active links: 1234 protocol: tcp CS−sensitivity: ani
BLUES=OFF
BLUES=ON
rel. dwell time [%]
0%
25%
50%
75%
100%
0
20
40
60
80
100
120
140
160
Time [s]
MAC states
tx−busy
rx−busy
idle
others

Are you egoistic or altruistic ?

Data Rate	**Thr.**[1] [MBits]	**Power** [mW]
6	6.0	1
9	8.8	1
12	11.7	1
18	15.3	12
24	4.7	19
36	2.1	19
48	0.0	19
54	0.0	19





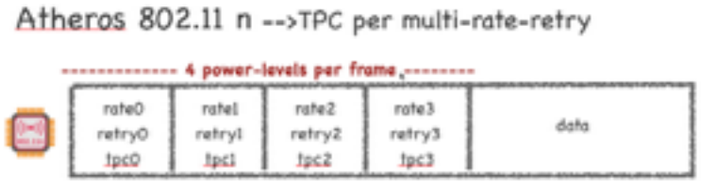
A joint decision based on preference is implemented !

my current work:



- get everything in upstream shape !!!

- Blues integration in Minstrel_HT





- since 3 weeks took a detour: enhancing Minstrel_HT

- my performance patches get sent in the next days



- Full fledged Minstrel-Blues ... this year

Thank You !